Energy fundamentals

Kinetic energy is the energy of motion. The formula is K = 0.5 * mass * speed squared. For mass 2 kg and speed 3 m/s, kinetic energy is 9 joules.

Gravitational potential energy

Near Earth's surface, gravitational potential energy is U = mass * g * height.

Use g = 9.8 m/s squared. A 1 kg book raised by 2 m gains 19.6 joules.